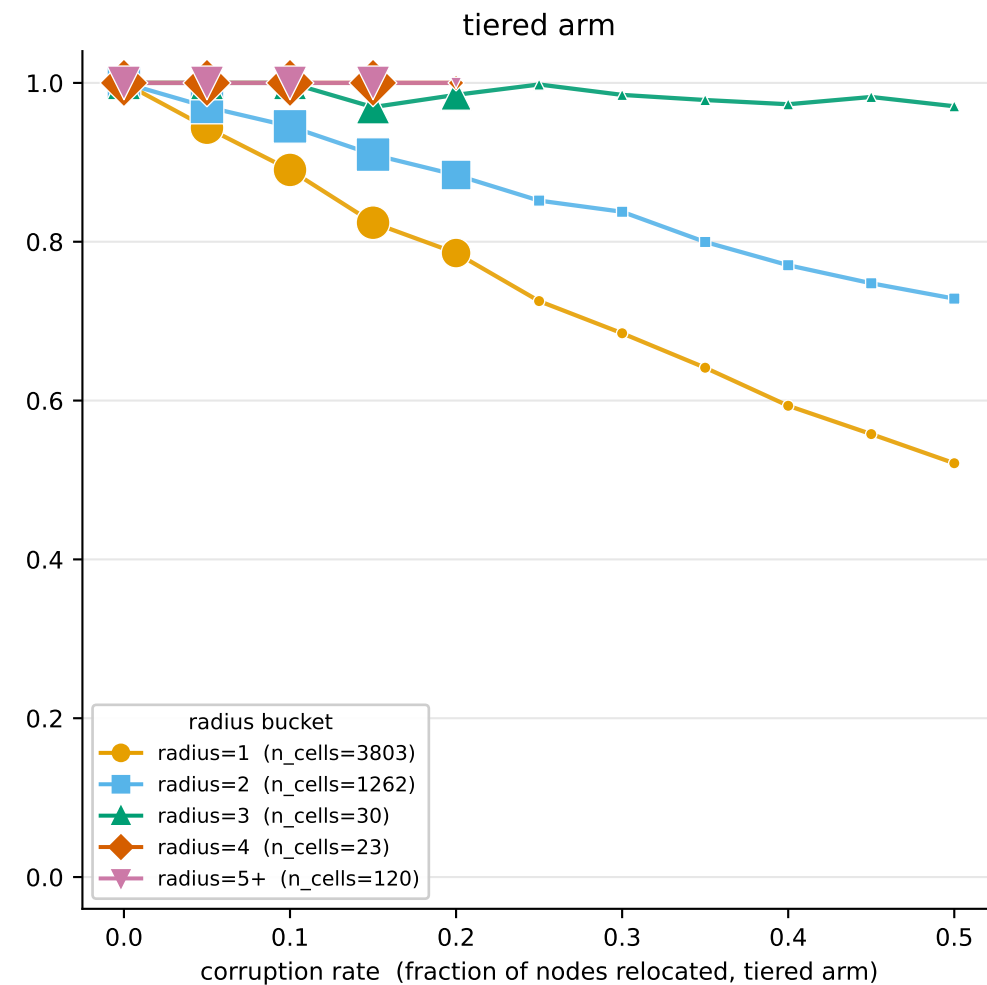
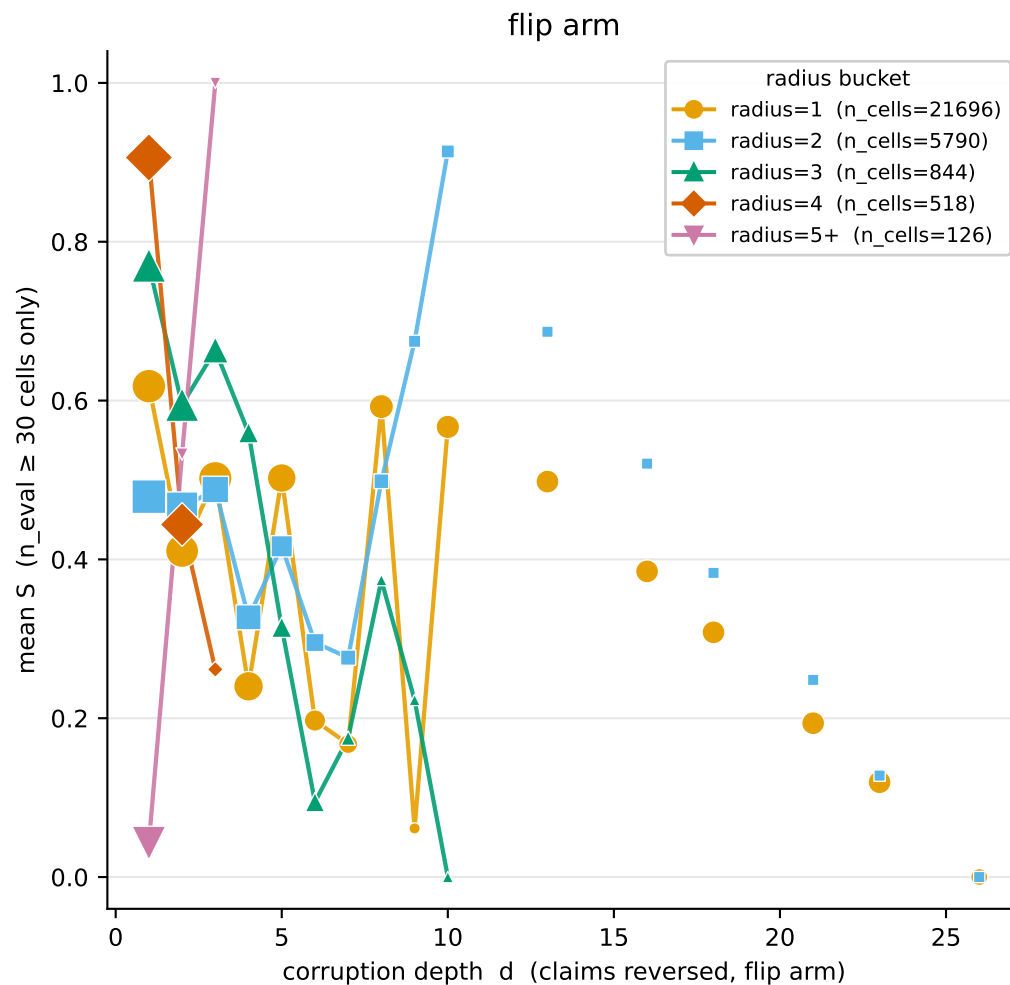


Survival by radius bucket, each arm on its own native axis



Shard cell files joined to analysis_units.csv on (shard_id, frame_row_id); pooled across all strata within each arm. Marker area $\propto \sqrt{(\text{contributing units})}$; a line breaks wherever the next grid point lacks a surviving cell (never bridged). r_{hop} assumes Conjecture 2 (hence Anti-Exchange Case B, verified not proved); the error is one-sided, so radii can only be too large.

Take: each arm's own native depth axis, radius-bucket lines pooled across strata; dropped cells break the line rather than being bridged.